# Session 4: Radiative Processes

Blackbody radiation, temperature scales, spectral lines

astro-lab – reading notes

**Status:** reading notes and exercise specification. The code for this session (`04_radiation/`) is not in the repository yet; the exercises below define what to build.

## Goals

Use the Planck function to relate a star's colour, temperature and luminosity, and fit an absorption line profile to extract a line centre, width and equivalent width.

## 1 The Planck function

The specific intensity of a blackbody at temperature $T$, per unit wavelength, is

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{\exp\!\left(\dfrac{hc}{\lambda k_B T}\right) - 1}.$$

Two limits are worth deriving yourself:

- **Rayleigh–Jeans** ($hc/\lambda k_B T \ll 1$): $B_\lambda \approx 2ck_BT/\lambda^4$.
- **Wien** ($hc/\lambda k_B T \gg 1$): $B_\lambda \approx (2hc^2/\lambda^5)\, e^{-hc/\lambda k_B T}$.

## 2 Wien and Stefan–Boltzmann

Setting $\mathrm{d}B_\lambda/\mathrm{d}\lambda = 0$ gives Wien's displacement law,

$$\lambda_{\max} T = b \approx 2.898 \times 10^{-3}\ \mathrm{m\,K}.$$

Integrating over all wavelengths and solid angle gives the flux $F = \sigma T^4$ with $\sigma \approx 5.670 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$. For a spherical star,

$$L = 4\pi R^2 \sigma T_{\mathrm{eff}}^4.$$

Using $L_\odot = 3.828 \times 10^{26}$ W and $R_\odot = 6.957 \times 10^{8}$ m gives $T_{\mathrm{eff},\odot} \approx 5772$ K, a good check for your code.

> **Why $L \propto R^2T^4$ matters.** A red giant and a white dwarf can share a colour but differ in luminosity by many orders of magnitude, purely through $R$. This is the geometric reason the HR diagram has distinct branches.

## 3 Spectral lines

Line profiles come from three broadening mechanisms:

- **Natural**: Lorentzian, from the finite lifetime of the level.
- **Doppler** (thermal and turbulent): Gaussian, $\sigma_D = \dfrac{\lambda_0}{c}\sqrt{\dfrac{k_B T}{m}}$ for thermal motion.
- **Pressure**: Lorentzian, from collisions.

Gaussian and Lorentzian combine into a **Voigt** profile (their convolution), usually evaluated with the Faddeeva function (`scipy.special.wofz`). The **equivalent width**

$$W_\lambda = \int \left(1 - \frac{F_\lambda}{F_c}\right) \mathrm{d}\lambda$$

is the width of a fully absorbed rectangle with the same area; it measures line strength independent of resolution.

## 4 Implementation plan

1. Write `planck(lam, T)` with units; test the Wien peak numerically against $b/T$ for $T = 3000$, 5772, 10000 K.
2. Integrate the Planck function numerically and check $\sigma T^4/\pi$ for the intensity integral.
3. Generate a synthetic spectrum: Planck continuum $\times$ a Voigt absorption line plus Gaussian noise.
4. Fit with `scipy.optimize.curve_fit`; report best-fit centre, width and uncertainties from the covariance matrix.
5. Compare the fitted line depth with the equivalent width computed by direct integration.

## 5 Exercises

1. Compute the peak wavelength for the Sun, Betelgeuse ($\sim 3600$ K) and Rigel ($\sim 12\,000$ K) and say which lie in the visible range.
2. Derive the Rayleigh–Jeans limit from the Planck function.
3. Estimate the thermal Doppler width of H$\alpha$ (656.28 nm) at 6000 K and compare with typical observed widths.
4. Show how the fitted width uncertainty scales with the noise level and with the number of pixels across the line.

## References


- Rybicki & Lightman, *Radiative Processes in Astrophysics*, ch. 1 and 10
- Gray, *The Observation and Analysis of Stellar Photospheres*
- Carroll & Ostlie, *An Introduction to Modern Astrophysics*, ch. 3 and 8